# Process-Aware Rewards and Reward Hacking: A Validated Replication of Zhang, Neubig & Yue (2025), §6, in the Contextual-Generalization Setting

Experiment 4 — DLT Final Project

Run on PACE ICE, 1× H100 (job 6589709); artifacts in `experiment4/`

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**Abstract**

Outcome-based reinforcement learning (RL) rewards for reasoning models are vulnerable to *reward hacking*: policies learn to emit correct final answers via invalid reasoning. Zhang, Neubig & Yue (2025) propose blending a process-verification signal into the reward and report that process-aware rewards raise strict (process-verified) accuracy and reduce hacking. We replicate this claim on the authors' released checkpoints in their contextual-generalization setting: one shared 100M-parameter base model and four GRPO runs differing *only* in reward composition, evaluated with $k$=64 samples on 2,000 problems (640k samples total). The paper's core claim is confirmed, and more decisively than its own presentation: the reward-hacking rate falls monotonically with process-supervision strength (52.3% → 45.8% → ≈42.3% → 40.5% on the long-tail context), with *every* step of the ladder significant under paired bootstrap after Bonferroni correction (strict vs. outcome: −5.31 pp [−6.37, −4.27]). Strict pass@1 gains from process rewards are decisive on edge-difficulty problems (+6.5 pp [+2.8, +10.3]) and directional but individually unresolved on the hardest band. During validation we additionally discovered a *letter-collision artifact* in the authors' released strict scorer that inflates absolute hacking rates (up to ∼6–8×) and deflates absolute strict pass@$k$ (12–22 pp) for all checkpoints equally. The headline process-vs-outcome contrast survives correction in full strength (−4.5 pp hacking, $p < 10^{-3}$); the finer ranking is verifier-regime-dependent — under corrected scoring the three process variants tie on hacking and the dense mixes dominate strict on accuracy.

## 1 Background and research questions

### 1.1 Task and evaluation formalism

Each reasoning problem is generated from a directed acyclic dependency graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ whose nodes are named quantities and whose edges encode functional dependencies $v_i = f_i\big((v_j)_{j \in \mathrm{pa}(i)}\big)$ with $f_i \in \{+, -, \times, \div\}$; difficulty is controlled by the operation count $\mathrm{op}(\mathcal{G}) = |\mathcal{E}|$. A contextual template $\tau$ renders $\mathcal{G}$ into natural language; the same graph family rendered under different templates yields structurally identical problems with different surface forms (*contexts*).

Given a model solution $s$, the evaluation parser deterministically extracts a predicted graph $\hat{\mathcal{G}} = (\hat{\mathcal{V}}, \hat{\mathcal{E}}, \widehat{\mathrm{val}})$ and final answer $\hat{a}$. For each gold node $v \in \mathcal{V}$ define the step-level correctness indicator

$$s(v; \hat{\mathcal{G}}, \mathcal{G}) \;=\; \begin{cases} 1, & v \in \hat{\mathcal{V}} \;\wedge\; \mathrm{pa}_{\hat{\mathcal{G}}}(v) = \mathrm{pa}_{\mathcal{G}}(v) \;\wedge\; \widehat{\mathrm{val}}(v) = \mathrm{val}(v), \\ 0, & \text{otherwise,} \end{cases} \tag{1}$$

the *process accuracy*

$$\text{ProcessAcc}(\hat{\mathcal{G}};\mathcal{G}) \;=\; \frac{1}{|\mathcal{V}|}\sum_{v\in\mathcal{V}} s(v;\hat{\mathcal{G}},\mathcal{G}), \tag{2}$$

and the *strict* (process-verified) correctness

$$\text{VerifiedCorrect}(\hat{a},\hat{\mathcal{G}}) \;=\; \mathbb{1}\Big[\text{ProcessAcc}(\hat{\mathcal{G}};\mathcal{G})=1\Big]\cdot\mathbb{1}[\hat{a}=a^*]\,. \tag{3}$$

Extra predicted nodes $\hat{\mathcal{V}}\setminus\mathcal{V}$ are permitted. All headline metrics use the strict criterion (3).

### 1.2 Reward designs under test

The paper's §6 augments the sparse outcome reward with process verification (its Eq. 3):

$$R \;=\; \alpha\, R_{\text{out}} \;+\; (1-\alpha)\, R_{\text{pv}}, \qquad R_{\text{out}}=\mathbb{1}[\hat{a}=a^*]\,, \quad R_{\text{pv}}=\text{ProcessAcc}, \tag{4}$$

and a stricter gated form

$$R_{\text{strict}} \;=\; \begin{cases} R_{\text{out}}, & \text{if } R_{\text{pv}}=1,\\ 0, & \text{otherwise.}\end{cases} \tag{5}$$

The five systems compared (all from one base model; the four RL runs used identical GRPO data, 90% context A / 10% context B, and differ only in $R$): **base** (no RL), **outcome** (answer-only; *verified* from the released training config, resolving an open question in the replication guide), **0.5/0.5** ($\alpha$=0.5; weights assumed — its config is absent from the release), **0.2/0.8** ($\alpha$=0.2; verified), and **strict** (Eq. 5; verified, `zero_on_process_mismatch`).

**Implementation divergence from Eq. (4).** The released reward code applies an *answer gate* to the mixed rewards: after computing $\alpha R_{\text{out}}+(1-\alpha)R_{\text{pv}}$ it zeroes the reward whenever the final answer is wrong. As trained, the mixes are therefore

$$R_{\text{mix}}^{\text{as-trained}} \;=\; \mathbb{1}[\hat{a}=a^*]\cdot\big(\alpha+(1-\alpha)\,R_{\text{pv}}\big), \tag{6}$$

i.e., the process term modulates reward magnitude *among answer-correct rollouts*; wrong-answer rollouts never receive the dense process signal that Eq. (4) implies. The literal ungated mix was never trained in the released family. Moreover, the training-time process reward used the same artifact-affected scorer of §3.4, so the released process-reward runs were themselves trained under a systematically noisy verifier.

### 1.3 Questions

**Q1** Do mixed/strict process rewards raise strict pass@$k$ over outcome-only? **Q2** Do they reduce the share of correct answers reached through invalid reasoning? **Q3** Do they shift errors away from structural failures (wrong dependencies, missing steps)?

## 2 Experimental setup

**Checkpoints.** HF `Interplay-LM-Reasoning/context_pretrain`, folder `idzoo_0.999zoo_0.001teacher/`: the base model is pre-trained on 99.9% context A (`crazy_zootopia`, op 2–20) + 0.1% context B (`teachers_in_school`, also spanning op 2–20 — this release is the paper's *appendix* A.6.1 "shared primitives" setting; the main-text §4 family with its op-2-only seed is the separate `context_pretrain_2` release); RL variants under `rl/contextzoo_0.9zoo_0.1teacher{,_process0.5,_process,_process_strict}`.

**Evaluation data.** `Interplay-LM-Reasoning/context` validation splits, op $\in \{12, 14, 16, 18, 20\}$, 200 problems per op, both contexts. Within a context, all five checkpoints answer byte-identical problems (verified), enabling paired analysis; the two contexts use different numeric instantiations, so cross-context comparisons are distribution-level. Difficulty bands: *edge* = op 12–14, *hard* = op 16–20.

**Sampling and budget.** Authors' vLLM evaluator; temperature 0.7, top-$p$ 1.0, $\leq$ 1,024 new tokens, $k = 64$ samples per problem: $5 \times 2 \times 1{,}000 \times 64 = 640{,}000$ samples (1 h 20 m wall time on one H100, ∼140 samples/s).

**Metrics.** For problem $i$ let $r_i \in \{0, \ldots, 63, \infty\}$ be the rank of the first strictly-correct sample. Then

$$\text{pass@}k \;=\; \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[r_i < k]\,, \tag{7}$$

with the answer-only variant defined analogously on $R_{\text{out}}$ alone. With $a_i = \sum_{j=1}^{64} \mathbb{1}[\hat{a}_{ij} = a_i^*]$ answer-correct samples and $s_i = \sum_{j=1}^{64} \text{VerifiedCorrect}_{ij}$ strictly-correct samples, the *reward-hacking rate* over a scope $S$ is the ratio of sums

$$H_S \;=\; \frac{\sum_{i\in S} \max(0,\, a_i - s_i)}{\sum_{i\in S} a_i} \;=\; \Pr\big(\text{invalid reasoning} \,\big|\, \text{correct answer}\big). \tag{8}$$

**Uncertainty.** Percentile bootstrap over *problems* ($B$=1,000 for per-variant CIs; problems are the independent sampling unit — the 64 draws within a problem are clustered and are never resampled individually). Because all variants share identical problems, headline *differences* use the paired bootstrap: resample problems once per replicate, compute the difference of the two variants' statistics on the same resample ($B$=5,000), and report the percentile interval; Bonferroni correction over the $m$=9 headline comparisons uses $\alpha = 0.05/m$.

**Validity.** Four independent audits (data integrity; clean-room metric recomputation, which matched our pipeline to $\sim 10^{-16}$ and the authors' own aggregator *exactly*; qualitative re-scoring of 240 samples, reproduced 240/240; paired statistical review) are reported in `analysis/validation/VALIDATION.md`. Zero parse failures, zero truncation at the token cap, zero duplicate samples.

# 3 Results

## 3.1 Q2 first: process supervision reduces reward hacking, monotonically

Table 1: Reward-hacking rate $H$ (Eq. 8), context B (teachers), overall, with 95% bootstrap CIs over problems; and paired differences vs. outcome-only. † survives Bonferroni ($\alpha = 0.05/9$).

| Reward | $H$ (%) | 95% CI | Paired Δ vs. outcome (pp) |
|---|---|---|---|
| base | 52.3 | [49.1, 55.7] | +6.51 [+4.74, +8.25]† |
| outcome | 45.8 | [42.7, 48.8] | — |
| 0.5/0.5 | 42.3 | [39.3, 45.7] | −3.48 [−4.35, −2.62]† |
| 0.2/0.8 | 42.2 | [38.7, 45.4] | −3.59 [−4.46, −2.73]† |
| **strict** | **40.5** | [37.3, 43.5] | **−5.31** [−6.37, −4.27]† |

Every step of the ladder base > outcome > mixes > strict is significant under the paired test and survives Bonferroni; strict is additionally below *both* mixes ($\approx -1.8$ pp, $p < 10^{-3}$). The ordering holds per band (hard: 62.1% → 44.3%) and, attenuated, on the well-covered zoo context (33.6% → 32.0%). **The paper's prediction that strict gating yields the lowest hacking rate is confirmed decisively**; this is the strongest result of the experiment.

### 3.2 Q1: strict pass@$k$ — decisive on edge, directional on hard

Table 2: Strict pass@1 / pass@64 (%), context B (teachers), by difficulty band.

| Reward | pass@1 edge | pass@1 hard | pass@64 edge | pass@64 hard |
|---|---|---|---|---|
| base | 33.5 | 11.5 | 80.8 | 50.7 |
| outcome | 44.8 | 30.3 | 81.0 | 60.7 |
| 0.5/0.5 | **51.2** | 32.5 | 81.5 | **62.8** |
| 0.2/0.8 | 47.8 | **33.5** | 80.5 | 59.7 |
| strict | 48.0 | 32.5 | **83.0** | 59.3 |

- *RL vs. base* is unambiguous: hard pass@1 +18.8 pp [+15.2, +22.5]; hard pass@64 +10.0 pp [+6.7, +13.3] (paired, Bonferroni-robust). RL transfers reasoning into the long-tail context given its 0.1% pre-training seed, consistent with the seed-dependence the paper establishes in §4/A.6.1.
- *Process vs. outcome, edge band*: 0.5/0.5 beats outcome-only by **+6.50 pp** [+2.75, +10.25] at pass@1 (significant, Bonferroni-robust).
- *Process vs. outcome, hard band*: all three process variants are ahead at pass@1 (+2.2 to +3.2 pp), sign-consistent across all five op counts (sign test $p = 0.031$), but no individual difference excludes zero under the paired test (best: 0.2/0.8, $p = 0.056$). The paper predicted ∼4–5 pp here; we observe the same direction at smaller magnitude. The best hard-band variant being 0.2/0.8 matches the paper's Fig. 7.
- *Zoo (well-covered context A)*: RL raises pass@1 (60.3 → ∼63%) but *lowers* edge pass@64 (94.5 → 89.8; paired base−outcome = +4.75 pp [+2.25, +7.25]) — RL sharpens existing ability and measurably narrows the sampled solution set where the base model was already strong, echoing the paper's §3 from an independent angle.

### 3.3 Q3: error decomposition

Table 3: Share of *all* samples per error category (%), teachers, hard band. Categories are non-exclusive except "correct".

| Reward | correct | wrong dep. | missing steps | wrong value | wrong answer |
|---|---|---|---|---|---|
| base | 11.7 | 86.5 | 46.6 | 79.3 | 69.2 |
| outcome | 29.5 | 68.1 | 30.1 | 47.8 | 40.5 |
| 0.5/0.5 | 32.0 | 66.7 | **24.8** | 47.8 | 40.0 |
| 0.2/0.8 | 32.0 | 66.8 | 27.6 | 48.2 | 40.5 |
| strict | 30.9 | 67.9 | 34.7 | 52.0 | 44.5 |

Process rewards shift errors away from structural failures as predicted — the dense mixes most clearly (missing steps 30.1% → 24.8%). Two nuances the paper's framing misses: (i) *strict*

*gating increases missing-step errors* relative to outcome-only (34.7% vs. 30.1%), plausibly because an all-or-nothing trace reward favors shorter, conservative solutions while dense per-step rewards directly pay for completing every step; (ii) among *genuinely* flawed solutions, base and strict error compositions are similar — the process-reward benefit is mostly *fewer* genuine errors rather than *different kinds*.

### 3.4 A measurement artifact in the released strict scorer

Validating per-sample verdicts by re-scoring solutions from scratch, we found a systematic defect in the authors' `SolutionParser`: its dependency extractor captures *every single letter* in an assignment's right-hand side, whose regex capture extends through trailing English prose. In the dominant `forwardreverse` problem mode the final step reads "$\ldots = 16$ *and we solve it through the following steps: Simplifying* $\ldots$", so letters from *words* become phantom dependencies whenever the model (or the gold solution) happened to assign those letters to earlier quantities. "Perfect" under Eq. (1) therefore partly requires the model's random variable-letter choices to collide with gold's phantom pattern. We verified the mechanism (structurally identical solutions flip from `dependency_mismatch` to perfect when the prose tail is stripped from both sides) and re-scored the full op $12 + 20$ population (25,600 samples/run):

Table 4: Original → artifact-corrected metrics (ops 12+20 population).

| Run | Hacking rate | Strict pass@1 |
|---|---|---|
| zoo/base | 29.8% → 4.6% | 64.8 → 85.0 |
| zoo/outcome | 29.0% → 4.2% | 68.5 → 90.5 |
| zoo/strict | 28.2% → 3.7% | 66.5 → 89.5 |
| teachers/base | 46.3% → 30.6% | 26.0 → 33.3 |
| teachers/outcome | 42.5% → 23.7% | 38.8 → 51.0 |
| teachers/strict | 37.5% → 19.6% | 40.3 → 53.5 |

Take-aways, updated after re-scoring the *full* population of all ten runs with the corrected instrument (`extensions/n0.4_corrected_scorer/`): (i) absolute hacking rates from this scorer are inflated (∼6–8× on zoo, ∼1.5× on teachers) and absolute strict pass@$k$ deflated by 12–22 pp. (ii) Under corrected scoring the decisive finding survives at full strength: **any process signal reduces hacking by ≈4.5 pp over outcome-only** (paired, $p < 10^{-3}$, teachers, all three variants). Two finer orderings are *revised*: strict's position as uniquely lowest-hacking does not survive — the three process variants statistically tie (all pairwise CIs cross zero) — and the dense mixes dominate strict on corrected pass@1 (+3.1 to +3.5 pp, $p < 0.02$) while strict's pass@1 advantage over outcome-only vanishes (+0.7 pp, n.s.); the zoo hacking ladder is entirely artifact. Because the training-time reward used this same defective verifier (§2), a natural hypothesis — testable with the released training stack — is that the strict gate, an all-or-nothing function of the noisy verifier, was the variant most distorted by training-time verifier noise. (iii) The paper used this same scorer, so its absolute Figs. 7–8 numbers likely carry the same artifact, and its strict/0.2–0.8 ranking is exactly the kind of conclusion this re-analysis shows to be verifier-regime-dependent; comparisons against the paper should be limited to the process-vs-outcome contrast. Genuine hacking remains real: audited corrected-hacked samples contain internally consistent but factually wrong derivations landing the right answer, and in 0/900 audited cases did the scorer claim a step missing that was present in the text.

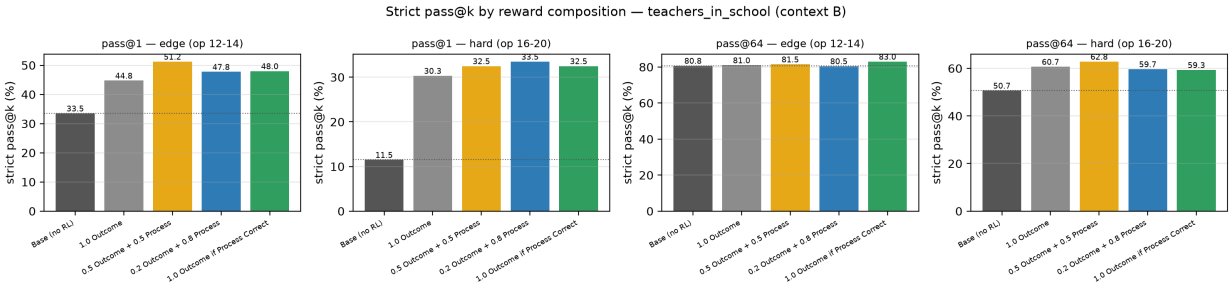



Figure 1: Strict pass@1/pass@64 by reward composition and band, context B (paper-Fig. 7 analogue). Dotted line: base model.

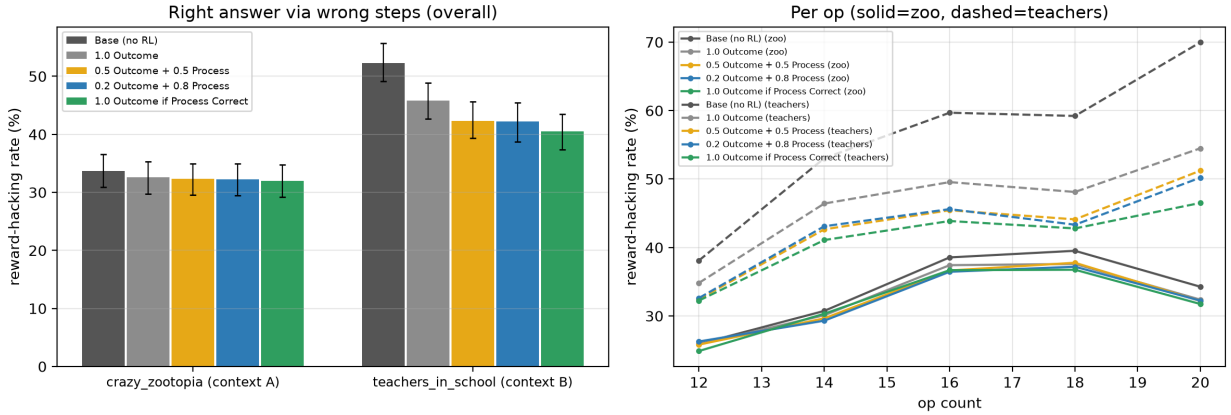



Figure 2: Reward-hacking rate (Eq. 8) by reward composition: overall per context (left, with 95% CIs) and per op count (right).

## 4 Conclusions vs. the paper's predictions

1. *"Any process signal raises strict* pass@1 *by ∼4–5 points over outcome-only on the hardest problems."* — **Directionally supported** on hard (+2.2 to +3.2 pp, sign-consistent, individually unresolved); **unambiguous on edge** (+6.5 pp [+2.8, +10.3]).

2. *"0.2/0.8 and strict are best; strict has the lowest hacking rate."* — **Confirmed under the released scorer** (0.2/0.8 best hard-band pass@1; hacking ladder Bonferroni-robust at every step). Under the *corrected* scorer the process-vs-outcome reduction survives in full (−4.5 pp, $p < 10^{-3}$) but strict's uniqueness does not: the three process variants tie on hacking and the dense mixes dominate strict on accuracy — the reward ranking is verifier-regime-dependent (§3.4).

3. *"Process rewards shift errors away from wrong dependencies and missing steps."* — **Supported in aggregate**, with the strict-gating missing-step nuance (§3.3) and the fewer-not-different-errors mechanism.

## 5 Limitations

One training run per reward variant (CIs quantify evaluation uncertainty only — seed variance cannot be excluded; the paper shares this limitation). The paper ran §6 on extrapolation; the released reward variants exist only in the context setting (directional comparability). The scorer artifact (§3.4) biases absolute strict metrics for all systems; we report orderings and paired differences, which are robust. The outcome checkpoint's reward is answer-only (verified from its released config); the 0.5/0.5 checkpoint's exact weights remain assumed. $k$=64 rather than the paper's 128, to fit a 3-hour single-GPU budget.

## 6 Reproducibility

One command on PACE (`sbatch scripts/run_all_pace.sbatch`); pinned environment (vLLM 0.10.1.1, transformers 4.55.2, torch 2.7.1). Raw per-sample outputs in `results/`; metrics, figures, and the four-audit validation (clean-room recomputation, paired statistics with seed 12345, 48 annotated qualitative cases, artifact re-scoring) in `analysis/`. Total compute: 1 h 20 m on one H100.